Chapter 5: Differentiation

Section: Taylor's Theorem

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

From the Mean Value Theorem to Taylor

- The mean value theorem (5.10), rewritten:

$$f(\beta) = \underbrace{f(\alpha)}_{\text{constant approximation}} + \underbrace{f'(x)(\beta - \alpha)}_{\text{exact error term}} \quad \text{for some } x \text{ between } \alpha, \beta$$

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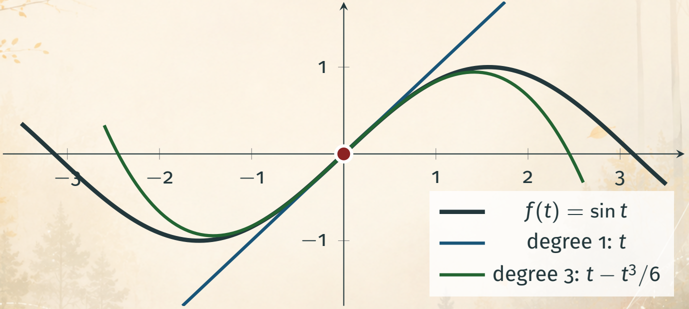
- With n derivatives available, we can do better: approximate f by a polynomial of degree $n-1$ built at α
- Taylor's theorem: the error keeps the *same shape* — one nth derivative at one unknown point: $\dfrac{f^{(n)}(x)}{n!}(\beta-\alpha)^n$

Taylor Polynomials Hug the Curve



Both polynomials are built *only* from derivatives of f at the red point $\alpha = 0$.

The Taylor Polynomial

The Setup for Theorem 5.15

Standing hypotheses

- f is a real function on $[a, b]$; n is a positive integer
- $f^{(n-1)}$ is continuous on $[a, b]$
- $f^{(n)}(t)$ exists for every $t \in (a, b)$

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Two distinct points

α, β are distinct points of $[a, b]$:

α = where we *know* f (center of expansion)

β = where we *want* f (target)

The Taylor Polynomial

The Taylor polynomial of f at α

$$P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t-\alpha)^k \tag{23}$$

Written out:

$$P(t) = f(\alpha) + f'(\alpha)(t-\alpha) + \frac{f''(\alpha)}{2!}(t-\alpha)^2 + \cdots + \frac{f^{(n-1)}(\alpha)}{(n-1)!}(t-\alpha)^{n-1}$$

Why P Is the Right Polynomial

P copies all available derivatives of f at α

$$P^{(k)}(\alpha) = f^{(k)}(\alpha) \qquad \text{for } k = 0, 1, \ldots, n-1$$

Reason: differentiating $\dfrac{(t-\alpha)^j}{j!}$ exactly k times and setting $t = \alpha$ gives 1 if $j = k$, and 0 otherwise. Each coefficient of P is tuned to match one derivative.

Taylor’s Theorem

Theorem 5.15: Taylor's Theorem

Theorem 5.15 (Taylor)

Under the standing hypotheses, there exists a point x between α and β such that

$$f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n. \tag{24}$$

Key idea: the value of f at β = polynomial prediction $+$ an exact error term of mean-value type.

The Case $n = 1$: The Mean Value Theorem

Taylor with $n = 1$

The sum (23) has the single term $k = 0$, so $P(t) = f(\alpha)$, and (24) reads

$$f(\beta) = f(\alpha) + f'(x)(\beta - \alpha).$$

This is exactly the mean value theorem (5.10).

Taylor's theorem = *the mean value theorem of order n.*

What the Theorem Buys Us: Error Estimates

From an unknown point to a usable bound

If $\left|f^{(n)}(t)\right| \leq M$ for all t between α and β, then (24) gives

$$\left|f(\beta) - P(\beta)\right| \;\leq\; \frac{M}{n!}\,|\beta - \alpha|^{n}.$$

- f can be computed to any accuracy from finitely many derivatives
- $n!$ grows extremely fast — the bound improves rapidly with n

Proof of Taylor’s Theorem

Proof Strategy

Goal: find x between α and β with $f(\beta) = P(\beta) + \dfrac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$.

Approach:

1. Let M be the number that makes $f(\beta) = P(\beta) + M(\beta - \alpha)^n$ true. Then show $n!\, M = f^{(n)}(x)$ for some x between α and β
2. Build the auxiliary function $g(t) = f(t) - P(t) - M(t - \alpha)^n$
3. Reduce the goal to: $g^{(n)}(x) = 0$ for some x
4. g vanishes n times over: repeated mean value theorem produces such an x

Technique: auxiliary function $+$ iterated mean value theorem.

Proof of Theorem 5.15 — Step 1: Define M and g

Step 1: Set up

Let M be the (unique) number defined by

$$f(\beta) = P(\beta) + M(\beta - \alpha)^n \tag{25}$$

and put

$$g(t) = f(t) - P(t) - M(t - \alpha)^n \qquad (a \leq t \leq b). \tag{26}$$

To show: $n!\, M = f^{(n)}(x)$ for some x between α and β.

Proof of Theorem 5.15 — Step 2: Reduce to a Zero of $g^{(n)}$

Step 2: Differentiate (26) n times

P has degree $\leq n-1$, so $P^{(n)} \equiv 0$; $\quad \left[(t-\alpha)^n\right]^{(n)} = n!$

$$g^{(n)}(t) = f^{(n)}(t) - n!\,M \qquad (a < t < b). \tag{27}$$

New goal: $g^{(n)}(x) = 0$ for some x between α and β.
Then (27) gives $n!\,M = f^{(n)}(x)$.

Proof of Theorem 5.15 — Step 3: g Vanishes Many Times

Step 3: Zeros of g and its derivatives

Since $P^{(k)}(\alpha) = f^{(k)}(\alpha)$ for $k = 0, \ldots, n-1$:

$$g(\alpha) = g'(\alpha) = \cdots = g^{(n-1)}(\alpha) = 0. \tag{28}$$

Moreover, the choice (25) of M gives $g(\beta) = 0$.

g vanishes at both α and β, and its first $n-1$ derivatives vanish at α.

Proof of Theorem 5.15 — Step 4: The Cascade

Step 4: First application of the mean value theorem

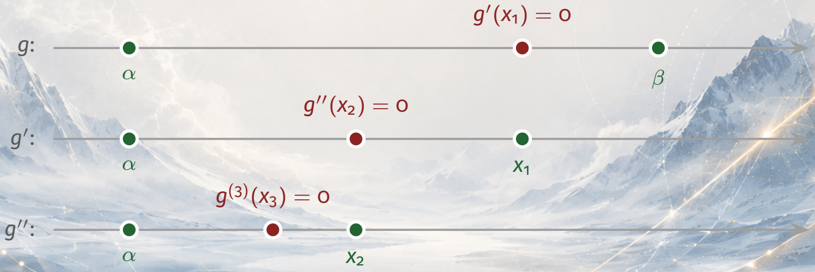
$g(\alpha) = g(\beta) = 0 \implies g'(x_1) = 0$ for some x_1 between α and β.

Proof of Theorem 5.15 — Step 4: The Cascade

Step 4: Iterate, using (28) at each stage

$g'(\alpha) = g'(x_1) = 0 \implies g''(x_2) = 0$ for some x_2 between α and x_1

… after n steps: $g^{(n)}(x_n) = 0$ for some x_n between α and x_{n-1}.



$\vdots$ (n steps in total)

Proof of Theorem 5.15 — Conclusion

Conclusion

Each x_k lies between α and x_{k-1}, hence between α and β. Take $x = x_n$:

$$g^{(n)}(x) = 0 \stackrel{(27)}{\Longrightarrow} n!\,M = f^{(n)}(x) \stackrel{(25)}{\Longrightarrow} \boxed{f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n}$$

This proves (24). ■

Summary

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Key takeaways from this section:

- **Theorem 5.15 (Taylor):** $f(\beta) = P(\beta) + \frac{f^{(n)}(x)}{n!}(\beta - \alpha)^n$ for some x between α and β, where $P(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!}(t - \alpha)^k$

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- **It generalizes the MVT** ($n = 1$), and a bound $|f^{(n)}| \leq M$ turns it into the error estimate $|f(\beta) - P(\beta)| \leq \frac{M}{n!}|\beta - \alpha|^n$

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- **It generalizes the MVT** ($n = 1$), and a bound $|f^{(n)}| \le M$ turns it into the error estimate $|f(\beta) - P(\beta)| \le \frac{M}{n!}|\beta - \alpha|^n$
- **Proof pattern:** auxiliary function $g = f - P - M(t - \alpha)^n$; and a cascade of n mean value theorems

Coming up next: Differentiation of Vector-Valued Functions — what survives (and what fails!) when f takes values in $\mathbb{R}^k$.